Table S1: Positioning against representative defense approaches. The comparison describes problem scope, not an empirical performance ranking.

Approach	Intervention or emphasis	Relationship to this study	Comparison boundary
Malware-oriented CPPS analysis [8], [9], [10]	Spreading, incubation, dependency, and protection rules	Supplies the malware and coupling motivation	Does not by itself validate this controller's state-continuation interface
Moving-target defense [11], [12]	Changes system characteristics to impede adversarial inference or attacks	Demonstrates established adaptive cyber defense	Intervention space differs from selecting protected cyber nodes
Adaptive nonlinear power-system defense [13]	Adaptive moving-target defense in a dynamic power-system setting	Recent comparator for adaptive-defense positioning	No matched numerical benchmark is claimed here
Dynamic-botnet resiliency [14]	Integrated cyber–physical resiliency under dynamic botnet attacks	Related cross-layer decision setting	Attack model and action space require alignment before numerical comparison
Present architecture	Rollout selection among interpretable protection sets with a baseline acceptance gate	Separates predicted gain, realized loss, and implementation contract	Service proxies and limited rolling validation; no global safety or superiority claim

Table S2: Reference parameters and sensitivity profiles. Values are design settings; retrospective sensitivity results do not establish an optimum.

Parameter	Value or profile	Role and qualification
Spreading probabilities	$\beta = .01, \alpha = .15, \delta = 10^{-5}$	Reference stealthy campaign; probabilities per simulation step
Protection multiplier	$\chi = .2$ in the rolling study	Reduces acquisition of infection, without curing existing E/I states
Objective weights	$(.68, .24, .06, .02)$	EP, PAS, normalized RT, and negative RI; reference preference profile
Degree-prioritized risk profile	$(.36, .18, .08, .08, .20, .10)$	Sensitivity comparator; order: degree, betweenness, power degree, capacity, infection, visibility
Distributed-feature risk profile	$(.20, .15, .15, .15, .20, .15)$	Comprehensive-library reference configuration; also evaluated in the risk-weight sensitivity study
Physical-impact weights	$(.35, .25, .20, .20)$	Load, capacity feature, degree, betweenness; service-importance proxy
Safe-Guard	$\gamma = .10, \kappa = .50$	Relative margin and uncertainty penalty; κ is not a confidence level
Rolling protocol	3 stages; 8 selection samples; 24 split-confirmation samples; $H \leq 32$	Independent supplemental implementation, distinct from the 20-rollout comprehensive-library scenario preview

Table S3: Topology-specific composite-score comparison. Gain is mean CDP EP minus mean adaptive EP; positive gain favors adaptation. Bootstrap intervals use the supplied topology audit.

Family	CDP EP	Adaptive EP	Absolute gain	95% interval	Relative gain
BA	.020609	.022541	-.001932	$[-.003335, -.000635]$	-9.37%
ER	.030505	.030822	-.000317	$[-.002516, .002135]$	-1.04%
WS	.046370	.032015	.014355	$[.011599, .017422]$	30.96%

Three graph seeds and 120 paired settings per family. These are results of the recorded supplemental protocol, separate from the policy-library rolling controller.

Table S4: Cross-scale composite-score gains and serial stage timing. Timing entries are seconds, mean / empirical P95 across eight configurations per size.

IEEE size	Confirmed EP gain vs CDP	Selection-stage time	Confirmation-stage time	Three-method wall mean
14	$+1.70\%$	1.280 / 1.869	2.421 / 3.207	3.787
57	-1.43%	3.577 / 4.588	4.254 / 5.951	7.988
118	-7.15%	4.524 / 7.413	5.859 / 9.131	10.571
300	-8.55%	12.136 / 16.661	13.603 / 20.088	26.042

Performance uses the cross-scale independently confirmed study; timings use the separate serial timing experiment, one replicate per configuration. The final column includes three methods and is not the latency of one engineering decision. The available timing manifest does not identify a complete CPU/hardware specification; these values are descriptive reference measurements, not a reproducible end-to-end real-time benchmark.

Table S5: Policy-library rolling component comparisons on the same rolling engine. Gain is PAS of reference B minus PAS of method A, so positive gain favors A.

Method A	Reference B	Mean PAS A	Mean PAS B	Gain [95% interval]	Holm p
FULL	CDP	.483687	.482881	$-.000806$ $[-.001815, .000198]$	.509
FULL	No dynamic risk	.483687	.483691	$.000003$ $[-.000421, .000610]$	1.000
FULL	No guard	.483687	.464499	$-.019188$ $[-.025131, -.014649]$	1.72×10^{-16}
FULL	Heuristic + guard	.483687	.483165	$-.000522$ $[-.001445, .000396]$	.588
Split-confirmation	FULL	.483230	.483687	$.000457$ $[-.000370, .001365]$	1.000
No guard	Heuristic, no guard	.464499	.506726	$.042227$ $[.037986, .046623]$	8.05×10^{-50}

400 paired setting–seed cases, fixed eight settings, three decision stages. "No guard" removes the acceptance gate but retains rollout selection. The heuristic comparisons hold guard presence fixed. Intervals and tests follow main manuscript Section V-C; positive-gain and low-p claims must be read in the sign convention above.

Table S6: Selected paired observation effects for IEEE118. Positive differences mean worse PAS under the perturbed observation.

Condition	Stress minus clean [95% CI]	Holm p
10-step delay	$-.005227$ $[-.010821, -.000260]$	1.000
30-step delay	$.007813$ $[.003597, .012406]$	.0537
20% missing labels	$-.004453$ $[-.010854, .001159]$	1.000
5% false positives	$.005428$ $[-.000086, .011526]$	1.000
Delay30 + missing20	$.006501$ $[.002440, .010622]$	.1589

The supplied analysis adjusts 20 stress comparisons across four sizes. Simulation steps have no assumed millisecond conversion. Mean-difference intervals and signed-rank tests concern different statistics.

Table S7: Corrective ACOPF scope and load-curtailment recovery.

Case	OPF profiles	Load scale	Shed MW
14	3/3	–	–
57	2/3	.908594	134.91
118	3/3	–	–
300	2/3	.896484	2912.95

Recovery entries apply only to the initially failed evening-peak profiles. Highest tested feasible uniform-load scales correspond to 9.1406% and 10.3516% curtailment. Every case has zero branches with nonplaceholder thermal ratings. A dash means recovery was not required for these tested profiles.

Table S8: Key notation and interpretation.

Symbol	Meaning
$\mathcal{G}^c, \mathcal{G}^p$	Communication and physical graphs
$\mathbf{M}, \mathcal{C}(u)$	Dependency matrix and buses coupled to cyber node u
$z, \bar{z}$	Generic node feature and its stated normalized value
B_d, Π, K	Protection budget, candidate library and candidate count
$\pi_k, \mathcal{U}_k$	Candidate policy and resulting protected node set
H, N_{cf}	Preview horizon and number of rollout samples
$\widehat{J}_k$	Estimated finite-horizon proxy objective
γ, κ	Relative switching margin and standard-error penalty
$\arg\min, \text{Top}_{B_d}$	Minimizing-candidate and largest-score selection operators
ϵ_J, ϵ_P	Hypothetical uniform errors required by the conditional inequalities

Table S9: Two-candidate local-information descriptive all-mode means and recorded controller cost. Each method has 9600 outcomes and 28800 decisions.

Method	Mean PAS	Mean stage computation (ms)
CDP	0.281039	0.235
CF_LOCAL	0.257409	54.053
CF_LOCAL_NO_GUARD	0.257410	17.350
CF_STATIC	0.271853	44.615
LOCAL_RANK	0.253506	0.213
STATIC_RANK	0.266549	0.215

Table S10: Empirical service-loss and baseline-relative risk in the matched multi-topology study.

Method	Mean PAS	P95	P99	ES95	Maximum	h_b (%)
CDP	0.281039	0.842478	0.914703	0.890313	0.976345	0.00
Direct static ranking	0.266549	0.791291	0.858478	0.832685	0.903361	33.67
Direct local ranking	0.253506	0.771157	0.846475	0.817491	0.907737	22.75
Static-candidate policy	0.271853	0.813252	0.893516	0.861866	0.976345	17.41
Local policy, selection only	0.257410	0.783566	0.879093	0.841292	0.976345	13.69
Guarded local policy	0.257409	0.784094	0.879093	0.841681	0.976345	13.55

Each method has 9600 paired contexts nested in 15 graphs. h_b is the fraction with PAS greater than the paired CDP outcome by more than 10^{-12}. ES95 is the mean of the worst 5% empirical probability mass; quantiles use linear interpolation. CDP has zero regret relative to itself by definition. These are descriptive risk summaries, not physical-safety certificates.

Table S11: Allocation-conditioned deterioration rate and overall baseline-relative regret.

Method	$n_{\neq b}$	$h_{b\|\neq b}$ (%)	Mean r_b^+	ES95 r_b^+
CDP	0	–	0.000000	0.000000
Direct static ranking	9600	33.67	0.008728	0.080762
Direct local ranking	9600	22.75	0.006201	0.075645
Static-candidate policy	5950	28.08	0.004332	0.059451
Local policy, selection only	7133	18.42	0.003868	0.060493
Guarded local policy	7106	18.31	0.003805	0.059749

$n_{\neq b}$ counts episodes with at least one executed protected-node set different from the CDP set at the decision state of that method. $h_{b|\neq b}$ uses those episodes as denominator. $r_b^+ = \max(\text{PAS}_m - \text{PAS}_b, 0)$ is positive paired regret. Regret summaries use all 9600 contexts; these episode measures do not identify individual-switch causal harm.

Table S12: Complete acceptance-margin and uncertainty-penalty replay grid.

γ	κ	Mean PAS	h_b (%)	P95	P99	ES95
CDP	–	0.037909	0.00	0.188005	0.378308	0.286322
0.00	0.00	0.046351	30.83	0.225770	0.460345	0.348629
0.00	0.25	0.044584	29.17	0.213528	0.390752	0.334428
0.00	0.50	0.042590	26.67	0.207520	0.371176	0.314430
0.00	0.75	0.042981	21.17	0.214240	0.390897	0.332163
0.00	1.00	0.040324	15.50	0.213528	0.380944	0.309433
0.05	0.00	0.046608	29.83	0.227534	0.460345	0.350439
0.05	0.25	0.044837	28.33	0.214240	0.391554	0.336691
0.05	0.50	0.042652	26.17	0.207520	0.371956	0.315734
0.05	0.75	0.042982	21.17	0.214240	0.390897	0.332163
0.05	1.00	0.040324	15.50	0.213528	0.380944	0.309433
0.10	0.00	0.045744	28.67	0.225479	0.391554	0.342489
0.10	0.25	0.044930	27.83	0.214240	0.391554	0.337252
0.10	0.50	0.042746	25.67	0.207520	0.371956	0.316296
0.10	0.75	0.043051	20.83	0.214240	0.390897	0.332163
0.10	1.00	0.040401	15.50	0.213528	0.380944	0.309433
0.15	0.00	0.045730	27.83	0.227534	0.391554	0.342693
0.15	0.25	0.045043	27.17	0.225479	0.391554	0.337518
0.15	0.50	0.042887	25.33	0.213432	0.371956	0.316960
0.15	0.75	0.043117	20.50	0.225479	0.390897	0.330936
0.15	1.00	0.040279	15.17	0.213528	0.380944	0.307005
0.20	0.00	0.043799	26.67	0.225479	0.382557	0.321389
0.20	0.25	0.043274	26.17	0.214114	0.371956	0.316152
0.20	0.50	0.042669	24.83	0.213432	0.371956	0.318348
0.20	0.75	0.042833	20.33	0.225479	0.390897	0.330936
0.20	1.00	0.039995	15.00	0.213432	0.380944	0.307001

Every cell uses the same 600 paired single-decision replay cases. The 25 cells reuse cases and are not independent replications. This comprehensive-library replay differs from the three-stage local-information study. All cells and all observed tail patterns are retained. The paired mean PAS gain relative to CDP is negative in all 25 cells; lower harm frequency under stricter settings therefore characterizes risk filtering, rather than universal mean dominance.

Table S13: All topology, observation and delay groups of the guarded local policy.

Graph	Observation	Delay	Mean gain vs CDP	h_b (%)	P95	P99
BA	Blind	0	+0.000000	0.00	0.899995	0.955611
BA	Blind	3	+0.000000	0.00	0.900323	0.955611
BA	Oracle	0	+0.064049	10.25	0.781747	0.831602
BA	Oracle	3	+0.060309	9.75	0.794469	0.834237
BA	50% recall	0	+0.021006	31.50	0.853929	0.883073
BA	50% recall	3	+0.018730	30.50	0.858506	0.885658
BA	80% recall	0	+0.043663	16.00	0.810253	0.853268
BA	80% recall	3	+0.041729	16.75	0.819084	0.865905
ER	Blind	0	+0.000000	0.00	0.684435	0.780111
ER	Blind	3	+0.000000	0.00	0.689445	0.780111
ER	Oracle	0	+0.048294	10.75	0.571615	0.643949
ER	Oracle	3	+0.047394	11.50	0.575317	0.654253
ER	50% recall	0	+0.016776	33.50	0.629301	0.718599
ER	50% recall	3	+0.017249	31.75	0.639132	0.720503
ER	80% recall	0	+0.033032	18.75	0.610902	0.677468
ER	80% recall	3	+0.032921	18.75	0.610960	0.683181
WS	Blind	0	+0.000000	0.00	0.281726	0.392362
WS	Blind	3	+0.000000	0.00	0.281726	0.392362
WS	Oracle	0	+0.027866	6.50	0.234031	0.322163
WS	Oracle	3	+0.026179	7.25	0.236892	0.322374
WS	50% recall	0	+0.013114	23.25	0.269443	0.350037
WS	50% recall	3	+0.012321	25.25	0.269443	0.350037
WS	80% recall	0	+0.021908	11.00	0.255640	0.329445
WS	80% recall	3	+0.020569	12.25	0.255640	0.329396

Each row contains 400 paired contexts nested in five graphs. Positive mean gain favors the guarded local policy. All 24 broad groups have nonnegative mean gain; this aggregation is not an all-configuration guarantee. At the finer 192-cell resolution, 143 means favor the policy, 48 blind controls tie and one cell has gain -0.001252 (BA, DDAC, budget20, onset90, 50% recall, delay3; 50 contexts). The complete finer-resolution CSV accompanies the source package.

Table S14: Complete fixed-graph independent-confirmation development sweep.

γ	κ	Mean PAS	h_b (%)	P95	P99	ES95
0.00	0.00	0.442661	12.50	0.939860	0.969039	0.963671
0.00	0.50	0.443493	10.42	0.939860	0.969039	0.963671
0.00	1.00	0.446600	10.42	0.939860	0.969039	0.963671
0.02	0.00	0.446410	10.42	0.948390	0.977752	0.973040
0.02	0.50	0.445817	8.33	0.948390	0.977752	0.973040
0.02	1.00	0.449167	8.33	0.948390	0.977752	0.973040
0.05	0.00	0.454536	8.33	0.948390	0.976630	0.972045
0.05	0.50	0.454536	8.33	0.948390	0.976630	0.972045
0.05	1.00	0.455727	8.33	0.948390	0.976630	0.972045
0.10	0.00	0.462592	2.08	0.948390	0.976630	0.972045
0.10	0.50	0.462592	2.08	0.948390	0.976630	0.972045
0.10	1.00	0.462592	2.08	0.948390	0.976630	0.972045

Each cell has 48 paired validation contexts on one fixed BA graph. ES95 uses the empirical probability-mass convention defined in this section. These reused development cases do not provide 12 independent test cohorts and are not the final multi-topology local-information evaluation. A separate 200-case holdout evaluates the configuration selected during development.